

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 6

Coordinate Geometry & Lines

Duration: 2.5 hours Total Marks: 100
Name: _____ Date: _____

SECTION A: Short Answer Questions (40 marks)

1. Find distance between (3, 4) and (0, 0) [2 marks]

2. Find midpoint of line joining (-2, 3) and (4, 9) [2 marks]

3. Find gradient of line through (1, 2) and (5, 10) [2 marks]

4. Find equation of line with gradient 2 through point (1, 3) [3 marks]

5. Two lines: $y = 2x + 1$ and $y = 2x + 5$. Are they parallel, perpendicular, or neither? [2 marks]

6. Find equation of circle with center (2, 3) and radius 5 [3 marks]

7. Find x-intercept and y-intercept of line $3x + 2y = 6$ [2 marks]

8. Check if point (1, 5) lies on line $y = 4x + 1$ [2 marks]

SECTION B: Long Answer Questions (60 marks)

9. Find equation of line perpendicular to $y = 2x - 3$ passing through $(2, 1)$. (4 marks)

10. A circle has equation $(x-3)^2 + (y+2)^2 = 16$. Find center and radius. (2 marks)

11. Find intersection points of $y = x^2 - 3x + 2$ and $y = x - 1$. (5 marks)

12. Line through $(1, 2)$ and $(4, 8)$. Find: (a) equation, (b) distance between points. (5 marks)

13. A triangle has vertices A(0, 0), B(4, 0), C(2, 3). Find: area, perimeter. (6 marks)

14. Reflection of point (3, 4) in line $y = x$. Find new coordinates. (3 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Distance between points

$$d = \sqrt{[(3-0)^2 + (4-0)^2]} = \sqrt{(9+16)} = \sqrt{25} = 5$$

Q2: Midpoint

$$M = ((-2+4)/2, (3+9)/2) = (1, 6)$$

Q3: Gradient

$$m = (10-2)/(5-1) = 8/4 = 2$$

Q4: Line equation

$$y - 3 = 2(x - 1), y = 2x + 1$$

Q5: Parallel/perpendicular

Both have gradient 2, so PARALLEL

Q6: Circle equation

$$(x-2)^2 + (y-3)^2 = 25$$

Q7: Intercepts

x-intercept: set $y=0$, $3x = 6$, $x=2$, y-intercept: set $x=0$, $2y=6$, $y=3$

Q8: Check if point on line

$y = 4(1) + 1 = 5$, so YES, (1,5) lies on line

SECTION B SOLUTIONS

Q9: Perpendicular line

Gradient of original = 2, perpendicular gradient = $-1/2$, $y - 1 = -1/2(x - 2)$, $y = -x/2 + 2$

Q10: Circle center and radius

Center (3, -2), radius = 4

Q11: Intersection points

$x^2 - 3x + 2 = x - 1$, $x^2 - 4x + 3 = 0$, $(x-1)(x-3) = 0$, $x = 1$ or 3 , $y = 0$ or 2 , points (1,0) and (3,2)

Q12: Line equation and distance

(a) gradient = $(8-2)/(4-1) = 2$, $y - 2 = 2(x - 1)$, $y = 2x$, (b) distance = $\sqrt{[(4-1)^2 + (8-2)^2]} =$

$$\sqrt{9+36} = \sqrt{45} = 3\sqrt{5}$$

Q13: Triangle area and perimeter

Area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 4 \times 3 = 6$, $AB = 4$, $AC = \sqrt{4+9} = \sqrt{13}$, $BC = \sqrt{[(4-2)^2 + (0-3)^2]} = \sqrt{13}$, Perimeter = $4 + 2\sqrt{13}$

Q14: Reflection in $y = x$

Swap coordinates: $(3,4)$ reflects to $(4,3)$